



REFERENCE

The smysl Document Format

Every construct of the .smy surface syntax, with a worked example for each

Vladimir Ulogov · 2026 · format smysl/0.1, kernel smysl.kernel/0.1

This reference covers one thing only: how to write a `.smy` document by hand — every record you can author, every field it takes, and the rules a validator will hold you to. For **why** the format is shaped this way, see `SMYSL_RATIONALE.typ`; this document only answers **how**. Every example below is real syntax, checked against the parser rather than invented for the page.

How a document is put together

A `.smy` file is a flat sequence of **records**, each starting at column 0. Order does not matter to the meaning of the document — a claim can cite evidence that appears later in the file — though it is conventional to write things in the order a reader would want to meet them. Blank lines separate records but carry no meaning of their own, and nothing in the format is a comment: every line is either part of a record or it is a parse error.

NO COMMENTS

There is no `//` or `#` comment syntax. A stray line that isn't part of a record's header, gist, body, or detail is a diagnostic (`SMY-E001`), not an ignored remark. If you want a note to a future reader, it has to be a real unit — typically `@prose`.

Four kinds of record are ever hand-authored in surface syntax:

CONSTRUCT	WHAT IT DECLARES
<code>@doc</code>	The document header — at most one per file, and optional.
a unit	<code>@claim</code> , <code>@evidence</code> , and the other thirteen kernel types — the thing being claimed, cited, or asked.
<code>@rel</code>	A typed edge between two units — <code>causes</code> , <code>rebut</code> , <code>warrant</code> , and eleven more.
<code>@thread</code>	A named, ordered, role-annotated walk over units — a brief, a narrative, an analysis.

Two more record kinds exist in the format — `attestation` and `contention` — but neither is something you write directly: an attestation is stamped on automatically by whatever tool ingests or authors a unit, and a contention is what `merge` produces when two documents disagree. Nothing below asks you to author either by hand.

The `@doc` header

If present, `@doc` must be the first thing in the file, and it fixes the document's own identity and the terms under which everything else in it should be read:

```
@doc smysl/0.1 {
  id: v/f1
  intent: incident-brief
  lang: en
  requires: ["smysl.kernel/0.1"]
  granularity: { profile: default }
  roots: [f/root-cause]
}
```

FIELD	DEFAULT	MEANING
<code>id</code>	<code>v/doc</code>	This document's own identifier — a view id, <code>v/<slug></code> .
<code>intent</code>	empty	A free-text label for what the document is for, e.g. <code>incident-brief</code> .
<code>lang</code>	<code>en</code>	A BCP-47 language tag — up to 35 characters, hyphen-separated alphanumeric segments.
<code>requires</code>	empty	Schemas a full-fidelity reader must implement: the kernel schema itself and any extensions.
<code>roots</code>	empty	The units this view is about — everything reachable from them is in the view; nothing is copied.
<code>threads</code>	empty	Thread ids this view publishes, if any.
<code>granularity</code>	<code>default</code> profile	How much one unit is allowed to say — see below.

A view is not a container: `roots` names starting points, and everything reachable from them via `deps`, `grounds`, and relation edges belongs to the view at zero copying cost. That is also what makes merging two documents a plain union rather than a conflict.

Granularity

`granularity` bounds how much a single unit is allowed to say, as three presets or a custom mix. Granularity constrains **production** — what you should write — not the store: a merged corpus mixing granularities is legal, not an error.

PROFILE	GIST (L0)	BODY (L1)	ADMISSION
coarse	≤ 30 tokens	120 – 400 tokens	topical — one unit may bundle a topic
default	≤ 30 tokens	40 – 120 tokens	single-assertion — one unit, one claim
fine	≤ 30 tokens	20 – 60 tokens	single-assertion

Under **single-assertion** admission — the default — a body that bundles more than one claim into a single unit is **SMY-E040**, an error, not a style complaint: split it into two units and a relation between them instead. A body outside its profile's range is only **SMY-W041**, a warning.

Units — the shared anatomy

Every unit, whatever its type, shares the same header and the same three-level text. Here is the fullest ordinary example the format has:

```
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait], salience:
0.8 }
~ The eu-west connection pool is saturated.
  Waits climbed from a steady 2 ms baseline to over 300 ms in the same hour the 4.2
  rollout reached eu-west, and no other shard's pool moved.

--
Per-shard breakdown at hourly resolution: eu-west crosses 250 ms at 09:14 UTC,
forty
minutes after the rollout's first canary step there. No other shard's pool exceeds
its usual 5 ms band over the same seven-day window.
```

FIELD	REQUIRED WHEN	WHAT IT IS
label	optional	The word right after the type, before <code>{</code> , e.g. <code>c/pool-saturation</code> — how the rest of the file refers to this unit. Not identity: two labels never make two units, and a unit needs no label at all if nothing else in the file must reference it by name.
status	always — defaults to speculative	How sure this unit is entitled to be — the six-rung ladder, next section.
gist (~ ...)	always	The one required line of text — L0. Must sit on the line immediately after the header, with no blank line between.
body	needed by derived / inferred gists that lean on it	L1 — a paragraph or two, the ordinary prose account.
detail	only after a body	L2 — unbounded, introduced by a lone <code>--</code> line. A detail with no body above it is SMY-E023 .
deps	when the gist needs context to parse	Other units this one's wording depends on to be understood — not evidence, just prerequisite reading.

FIELD	REQUIRED WHEN	WHAT IT IS
grounds	required by derived / inferred	Other units this one's truth rests on. Retract one and this unit's status is re-opened.
source	required by measured / cited	Where this unit grounds out externally – see below.
salience	optional	An authored override in [0,1] , clamped and quantised to 1/1024. Left unset, salience is computed rather than declared.

The gist must immediately follow the header – not even a blank line may separate them (SMY-E021). A gist continuation line is indented by exactly two spaces; a body or detail line is not indented at all.

Status: the six-rung ladder

STATUS	NEEDS	MEANING
unfounded	–	Reachable only by retraction; MUST NOT be authored (SMY-E034).
speculative	neither	Offered, not yet grounded – the default if you omit status .
inferred	grounds	A model reasoned it out from other units.
derived	grounds	Computed from other units by a deterministic procedure.
cited	source	A source you can open says so.
measured	source	An instrument recorded it.

Two mechanical rules sit on top of this table. **Rule M**: a unit's status can never outrank the weakest thing in its own **grounds** (SMY-E030) – a **derived** claim resting on a **speculative** one is itself no better than speculative, whatever status you write. **Rule T**: the **rung** an agent is working from caps what status it may even attempt:

RUNG	CEILING	WHAT IT IS
computed	derived	A deterministic tool, calculation, or parser.
document	cited	A user-supplied document or dataset.
web	cited	Fetches content, gated.
model	inferred	The model's own parametric knowledge – never higher, however confidently phrased.

No rung reaches **measured** – only an **imported** reading from an instrument may claim it; an authored unit marked **measured** is SMY-W035 at best.

Source references

```
source: { kind: metric, ref: "pool.wait_ms{shard=eu-west}", captured: 2026-07-09 }
```

kind is one of five, **ref** (or the longer spelling **reference**) is a free-text pointer whose shape depends on the kind, and **captured** is an optional YYYY-MM-DD date.

KIND	TYPICAL REF	CAPTURE DATE
url	a web address	Required – a fetched page is unverifiable later without one.

KIND	TYPICAL REF	CAPTURE DATE
file	a path or filename	optional
metric	a metric name or query	optional
tool	a tool or command name	optional
doc	a document reference or anchor	optional

The thirteen unit types

`KernelType` actually enumerates fifteen names, but two of them — `contention` and `packinfo` — name records that tooling produces (`merge`, `pack`); nothing below shows them being hand-authored, because in ordinary use they never are. The other thirteen are yours to write. Each example is complete and independently valid.

TYPE	WHAT IT RECORDS
claim	An assertion about the world.
evidence	A reading, instrument, or observation offered to ground a claim.
definition	What a term means, fixed so later claims can rely on it.
question	An open prompt the corpus has not yet answered.
hypothesis	A candidate answer, offered before it is tested.
finding	A conclusion resting on other claims.
procedure	A course of action — a “what to do,” not a “what is.”
decision	A commitment a person or team made.
constraint	A rule the rest of the reasoning has to respect.
observation	A plain record of what was seen.
data	A pointer to a dataset, rather than one reading.
artifact-ref	A pointer to something outside the document — a dashboard, a repo.
prose	Free-standing text that fits none of the above — still a first-class, citable unit.

```
@evidence e/pool-wait { status: measured, source: { kind: metric, ref:
"pool.wait_ms{shard=eu-west}", captured: 2026-07-09 } }
~ Pool acquisition wait rose from 2 ms to 310 ms over the same window.
```

An instrument reading grounds out externally, so `measured` demands a `source` — here a metric reference with the date it was captured.

```
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait] }
~ The eu-west connection pool is saturated.
```

A model reasoned this from the evidence above; `inferred` demands `grounds`, and rule M means this claim can never outrank `e/pool-wait` even if the wording sounds certain.

```
@definition d/p95 { status: cited, source: { kind: doc, ref: "sre-
handbook#latency" } }
~ The 95th percentile of request latency over a one-minute window.
```

A definition is grounded in a document a reader can open, not the author’s own say-so — without it, “tripled” in a later claim would have no fixed meaning.

```
@question q/root-cause { status: speculative }
~ What actually caused the eu-west latency spike?
```

A question needs neither `source` nor `grounds` — `speculative` is the floor, an open prompt nothing has answered yet.

```
@hypothesis h/pool-exhaustion { status: speculative }
~ Connection-pool exhaustion under the 4.2 rollout is driving the p95 rise.
```

A candidate answer to the question above, offered before it is tested against evidence.

```
@finding f/root-cause { status: inferred, grounds: [c/pool-saturation, c/
regression] }
~ Pool saturation is the leading cause but is not consistent with the canary.
```

A finding rests on other claims rather than on raw evidence; rule M caps it at the weakest of the two grounds it names.

```
@procedure p/rollback { status: speculative }
~ Roll the eu-west pool config back to the 4.1 settings and re-run the canary.
```

A course of action, not a fact — still a checkable unit, just one whose content is a “what to do.”

```
@decision de/rollback-approved { status: inferred, grounds: [f/root-cause] }
~ Approved: roll back the eu-west pool config tonight.
```

A commitment, grounded in whatever finding justified it — retract the finding and this decision’s grounding is reopened along with it.

```
@constraint co/no-friday-deploys { status: cited, source: { kind: doc, ref: "ops-
runbook#change-freeze" } }
~ No production rollbacks after 16:00 on a Friday.
```

A rule the rest of the reasoning has to respect, sourced from wherever that rule is actually written down.

```
@observation ob/canary-quiet { status: measured, source: { kind: metric, ref:
"canary.p95" } }
~ The 4.2 canary showed no latency regression over the same window.
```

A plain record of what was seen — it doesn’t ground anything by itself yet, but it can be cited later, exactly like any other unit.

```
@data da/latency-series { status: measured, source: { kind: file, ref: "auth-
p95-2026-07.csv" } }
~ Hourly p95 auth latency, all shards, 2026-07-02 through 2026-07-09.
```

Points at a dataset rather than asserting a single reading — the source is the file itself.

```
@artifact-ref ar/dashboard { status: cited, source: { kind: url, ref: "https://
grafana.internal/d/api-latency", captured: 2026-07-09 } }
~ The latency dashboard the on-call team watches.
```

Points at something that exists outside the document. Its source kind is `url`, the one kind where `captured` is mandatory rather than optional.

```
@prose pr/note { status: speculative }
~ Worth a follow-up: check whether the 4.1 pool size was ever tuned for eu-west's
traffic shape.
```

The escape hatch — a citable aside that doesn't fit any more specific type.

NAMESPACE CONVENTION, NOT GRAMMAR

`e/`, `c/`, `d/`, `f/`, and the rest above are a **habit**, not a rule. A label is any `ident/ident` pair — lowercase letters, digits, `-`, `_` — and nothing in the grammar reserves any prefix for any unit type. Pick short, memorable prefixes and stay consistent within a file; the parser does not care what they are.

Relations (`@rel`)

A relation is a typed, directed edge between two units, written on one line:

```
<from> --<kind>--> <to> [{ weight: <0..1>, note: <ref> }]
```

```
@rel c/canary-clean --rebuts--> c/pool-saturation { weight: 0.6 }
```

A direct objection. `rebuts` is the one kind rule R pins: wherever `c/pool-saturation` survives into a packed or rendered view, this edge — and the unit it points from — travels with it. `weight` is an optional strength for the rebuttal itself, clamped to `[0,1]`.

```
@rel c/pool-saturation --causes--> c/regression
```

Says the saturation claim is what produced the regression. `causes` is one of three **ordering** kinds (with `enables` and `sequences`) that a thread's automatic ordering can sort by, and one of two **support-bearing** kinds (with `answers`) that carry salience rank forward through the graph.

```
@rel d/p95 --warrant--> c/regression
```

Says the definition licenses reading the claim the way it is read — without it, “tripled” has no fixed meaning to check.

```
@rel c/pool-saturation-v2 --supersedes--> c/pool-saturation
```

Marks the newer claim as replacing the older one outright. **supersedes** and **retracts** are the two **lifecycle** kinds — the only ones that change what the corpus should currently be read as believing, rather than adding a new relationship alongside what was already there.

```
@rel c/pool-saturation --x.sre/mitigates--> p/rollback
```

An unrecognised kind — anything shaped `x.<domain>/<kind>` — is kept and stays routable rather than dropped (SMY-W013); a reader without the `x.sre` extension treats it as a plain **elaborates** edge instead of losing it.

The fourteen kernel kinds, for reference:

KIND	READS AS
elaborates	adds detail without changing the claim
contrasts	sets two things against each other
concedes	grants a point while maintaining the larger claim
causes	one thing produced another (ordering , support-bearing)
enables	one thing made another possible without directly producing it (ordering)
exemplifies	gives a concrete instance of a general claim
conditions	states an “only if” dependency
sequences	orders two things in time, with no causal claim (ordering)
answers	resolves a question (support-bearing)
rebutts	directly objects to the target — pinned by rule R
warrant	licenses an inference or a reading
backs	supports a warrant
supersedes	replaces the target outright (lifecycle)
retracts	withdraws the target outright (lifecycle)

Threads (`@thread`)

A thread is a named, ordered, role-annotated walk over units — the same units the corpus already has, arranged into a reading order for one audience. Its header takes a **schema** (which fixes the allowed roles), an **owner** (an agent id — a thread is **owned** state, last- writer-wins **per owner**, so two agents publishing the same thread id never conflict), and a gist. Steps follow, indented by two spaces:

```
<role> → <unit>[: <note>]
```

Each of the five schemas fixes a different sequence of roles. One worked example per schema:

```
@thread t/brief { schema: brief, owner: "human:vladimir" }
~ Auth p95 tripled in eu-west; pool saturation is leading but contested.
  bottom-line → f/root-cause
```

```
support → c/pool-saturation
risk → c/canary-clean
```

brief — **bottom-line, support, risk, ask** — walks a reader from the one-line verdict down to what backs it and what argues against it. Not every role has to be used.

```
@thread t/root-cause-qa { schema: qa, owner: "human:vladimir" }
~ What caused the latency spike, and how sure are we?
question → q/root-cause
evidence → e/pool-wait
answer → f/root-cause
caveat → c/canary-clean
```

qa — **question, evidence, answer, caveat** — is literally a question answered in public, with its evidence and its own best objection attached.

```
@thread t/incident-story { schema: narrative, owner: "human:vladimir" }
~ How a quiet Tuesday became an eu-west incident.
setup → ob/canary-quiet
complication → e/pool-wait
turn → c/pool-saturation
resolution → de/rollback-approved
coda → f/root-cause
```

narrative — **setup, complication, turn, resolution, coda** — orders the same kind of units as a story, for an audience that needs to follow how the conclusion was reached.

```
@thread t/deep-dive { schema: analysis, owner: "model:anthropic/claude-opus-5" }
~ Full trace of the eu-west pool-saturation analysis.
context → ob/canary-quiet
tension → e/pool-wait
approach → h/pool-exhaustion
finding → c/pool-saturation
rebuttal → c/canary-clean
implication → f/root-cause
next → p/rollback
```

analysis — **context, tension, approach, finding, rebuttal, implication, next** — is the longest schema, a full research trace with its own slot for what to do next.

```
@thread t/rollback-plan { schema: plan, owner: "human:vladimir" }
~ Rolling eu-west back to 4.1 tonight.
goal → de/rollback-approved
constraint → co/no-friday-deploys
step → p/rollback
risk → c/canary-clean
```

plan — **goal, constraint, step, decision, risk** — orders the units that make up a course of action rather than an argument.

ARROWS AND NOTES

Either `→` (U+2192) or the ASCII `->` is accepted when **reading**; the canonical writer (`smysl fmt`) always emits `→`. A step may carry a trailing `: note` after the reference, as free text — support `→ c/pool-saturation: the strongest single reading`.

`risk` appears in both `brief` and `plan` — it is one role shared between schemas, not two separate ones. A step whose role the thread's schema does not declare is not rejected; it is reported as a **foreign role**, kept rather than dropped.

Identifiers, at a glance

KIND	SHAPE	EXAMPLE
Label	<code>ident/ident</code> , lowercase	<code>c/pool-saturation</code>
Thread / view / contention id	same shape, label-typed	<code>t/brief</code> , <code>v/f1</code> , <code>k/pool-vs-index</code>
Agent id	<code>(model\ human\ tool):name[/tag]</code>	<code>model:anthropic/claude-opus-5</code> , <code>human:vladimir</code>
Kernel type	a bare word	<code>claim</code> , <code>evidence</code>
Kernel schema	<code>smysl.kernel/MAJOR[.MINOR]</code>	<code>smysl.kernel/0.1</code>
Extension schema	<code>x.<domain>/<segment></code>	<code>x.sre/incident</code>
Language tag	BCP-47-shaped, ≤ 35 chars	<code>en</code> , <code>en-GB</code> , <code>zh-Hans-CN</code>
Content hash (uid)	<code>b3:</code> + base32, machine-facing	<code>b3:cvhirtgs2mpvli2ethhyeo32uf...</code>

You will essentially never type a `b3:` hash by hand — that is exactly what labels are for. A hash is what a unit's identity actually is; a label is the name you gave it so you never have to write the hash yourself. An agent id's kind is fixed to one of three words; a model id's segment after the first `/` groups corroboration by provider, which is why `model:anthropic/claude-opus-5` and `model:anthropic/claude-haiku-4-5` still corroborate as the same provider.

Validation, at a glance

A parse never simply fails: a malformed record becomes a diagnostic with a byte span, and parsing recovers at the next record start, so one bad unit costs you that unit, not the whole file. The codes below are the ones ordinary hand-authoring runs into.

CODE	WHAT IT MEANS
SMY-E001	General surface parse error — a malformed header, an unknown type, a stray line.
SMY-E020	Body or detail names a unit's identifier that isn't listed in <code>deps</code> or <code>grounds</code> .
SMY-E021	Missing gist, or the gist doesn't immediately follow the header.
SMY-E022	Gist exceeds the profile's <code>l0_max</code> .
SMY-E023	<code>detail</code> present without a <code>body</code> above it.

CODE	WHAT IT MEANS
SMY-E030	Rule M — status exceeds the weakest of its own grounds.
SMY-E031	<code>derived</code> / <code>inferred</code> with empty <code>grounds</code> .
SMY-E032	<code>measured</code> / <code>cited</code> without a <code>source</code> .
SMY-E033	Rule T — status exceeds the ceiling for the authoring rung.
SMY-E034	<code>unfounded</code> authored directly — only reachable by retraction.
SMY-E040	More than one assertion in a body under <code>single-assertion</code> admission.
SMY-E060	A label referenced but never defined anywhere in the document.
SMY-E061	A cycle in <code>deps</code> or <code>grounds</code> .
SMY-W013	Unknown relation kind — kept, treated as <code>elaborates</code> for closure.
SMY-W035	<code>measured</code> on an authored (not imported) unit.
SMY-W041	Body length outside the profile's <code>ll_range</code> — advisory only.

AT A GLANCE

- Four constructs are hand-authored: `@doc` once, units as many as you need, `@rel` for typed edges, `@thread` to walk them in an order.
- Every unit shares one anatomy — label, status, gist/body/detail, `deps` / `grounds`, `source`, `salience` — whatever its type.
- Status only ever falls, never rises, across `grounds` (rule M) and is ceilinged by the rung doing the authoring (rule T); only an import may write `measured`.
- Thirteen unit types are yours to write; `contention` and `packinfo` are tooling-only.
- Fourteen relation kinds, five thread schemas, twenty-four roles — all closed sets, all with an escape hatch (`x.<domain>/<kind>`) that degrades gracefully rather than being dropped.
- There are no comments, no blank line between a header and its gist, and no reserved label namespaces — only what the grammar actually checks.